DAMIEN CHEMISTRY OLYMPIAD NOTES

Chemical Bonding, Molecular Structure and Intermolecular Forces

Complete Question Bank with Worked Answers

Contents

A. Nature of Bonding 1
1. All Chemical Bonding as Electrostatic Attraction 1
2. Ionic, Covalent, Coordinate and Metallic Bonding 1
3. Electron Transfer Versus the Resulting Ionic Attraction 2
4. Delocalisation of Electrons 2
5. Dot-and-Cross Diagrams 3
6. Formal Charge 4
7. Resonance 4
B. Orbital Overlap 5
8. Formation of σ and π Bonds 5
9. Relative Strength of σ and π Overlap 5
10. Restricted Rotation Around Double Bonds 6
11. Bond Order, Bond Length and Bond Strength 6
12. Bond Polarity and Reactivity 7
C. Molecular Geometry 7
13. Electron-Domain Repulsion 7
14. Bonding Pairs Versus Lone Pairs 8
15. Relative Strengths of Electron-Pair Repulsions 9
16. Shapes and Approximate Bond Angles 9
17. Effects of Multiple Bonds on Geometry 10
18. Electron-Pair Geometry Versus Molecular Geometry 10
D. Molecular Polarity 11
19. Electronegativity and Bond Dipoles 11
20. Vector Cancellation of Bond Dipoles 11
21. Polar Bonds in Non-Polar Molecules 12
22. Molecular Symmetry 12
23. Effect of Lone Pairs on Polarity 13
E. Intermolecular Forces 13

24. Instantaneous Dipole–Induced Dipole Forces 13
25. Permanent Dipole–Permanent Dipole Forces 14
26. Hydrogen Bonding 15
27. Polarisability 15
28. Molecular Size, Surface Area and Shape 16
29. Intermolecular Versus Intramolecular Bonding 16
30. Solubility and "Like Dissolves Like" 17
F. Structures of Solids 17
31. Ionic Lattices 17
32. Metallic Lattices 18
33. Simple Molecular Solids 19
34. Giant Covalent Structures 19
35. Hydrogen-Bonded Solids 20
36. Structure–Property Relationships 20
37. Graphite, Diamond and Unfamiliar Allotropes 21
G. Integrated Problem Types 21
38. Boiling-Point Ranking 22
39. Structure from Physical-Property Data 22
40. Comparing Isomers 23
41. Identifying All Intermolecular Forces 23
42. Predicting Polarity from Unfamiliar Structures 24
43. Anomalous Properties of Water and Ice 24
44. Bond-Length and Bond-Energy Comparisons 25
45. Testing Proposed Lewis Structures and Shapes 25
Olympiad-Style Extensions 26
Extension 1: Counting Particles in a Unit Cell 26
Extension 2: Identifying a Unit Cell Using Density 27
Extension 3: Interpreting Bond Potential-Energy Curves 29
Extension 4: Interpreting a Boiling-Point Graph 30
Extension 5: Inferring Structure from Conductivity Data 32

A. Nature of Bonding

1. All Chemical Bonding as Electrostatic Attraction

1A Easy

Explain how the covalent bond in H_2 can be described entirely in terms of electrostatic attractions and repulsions.

Answer

Each electron is attracted electrostatically to both positively charged nuclei. These electron–nucleus attractions lower the energy of the system, while nucleus–nucleus and electron–electron repulsions oppose them. At the equilibrium bond length, the balance of these interactions gives a minimum in potential energy.

1B Medium

Compare the electrostatic attractions responsible for bonding in NaCl, H_2 and magnesium metal. State what particles are attracting one another in each case.

Answer

NaCl: attraction between Na^+ and Cl^- ions throughout an ionic lattice. H_2: attraction between both nuclei and the shared electron density between them. Magnesium: attraction between positive metal ions and delocalised valence electrons.

1C Hard

A student says, “Ionic bonds are electrostatic, whereas covalent bonds arise from electron sharing rather than electrostatic forces.” Evaluate this statement, accounting for electron–nucleus, nucleus–nucleus and electron–electron interactions.

Answer

The statement is misleading. Both ionic and covalent bonding are fundamentally electrostatic. In a covalent bond, sharing allows electron density to accumulate between nuclei, increasing attractive electron–nucleus interactions. Bond formation occurs only when these attractions outweigh nucleus–nucleus and electron–electron repulsions at an appropriate internuclear distance.

2. Ionic, Covalent, Coordinate and Metallic Bonding

2A Easy

Identify the principal type of bonding in Na_2O, Cl_2, NH_4^+ and solid copper.

Answer

Na_2O: ionic. Cl_2: covalent. NH_4^+ : covalent; one bond originates through coordinate-bond formation. Copper: metallic.

2B Medium

Explain the formation of the coordinate bond in NH_4^+ . Why are the four N–H bonds indistinguishable after the ion has formed?

Answer

NH_3 donates its nitrogen lone pair to H^+ , forming an N–H bond in which both bonding electrons originally came from nitrogen. Once NH_4^+ has formed, all four N–H bonds have equivalent electron density, length and strength. A coordinate bond is not a permanently different type of covalent bond.

2C Hard

Describe all the important types of bonding present in $[Cu(NH_3)_4]SO_4$, distinguishing between bonding within the complex ion, within ammonia and between the ions in the solid.

Answer

There are covalent N–H bonds within each NH_3 ligand, coordinate Cu–N bonds formed by donation of nitrogen lone pairs to Cu^{2+} , covalent S–O bonding with resonance within SO_4^{2-} , and ionic attraction between $[Cu(NH_3)_4]^{2+}$ and SO_4^{2-} in the solid lattice.

3. Electron Transfer Versus the Resulting Ionic Attraction

3A Easy

Distinguish between the transfer of an electron from sodium to chlorine and the ionic bond present in solid NaCl.

Answer

Electron transfer produces Na^+ and Cl^- ions. The ionic bond is the subsequent electrostatic attraction between oppositely charged ions. Electron transfer is therefore a step in ion formation, not the ionic bond itself.

3B Medium

Electron transfer from gaseous sodium to gaseous chlorine involves both endothermic and exothermic steps. Explain why the formation of solid NaCl can nevertheless be strongly exothermic.

Answer

Energy is required to atomise sodium and ionise Na atoms, while energy is released when Cl gains an electron. A large amount of energy is then released when gaseous Na^+ and Cl^- ions form the strongly attracting three-dimensional ionic lattice. This lattice formation can make the overall process exothermic.

3C Hard

A diagram shows one Na^+ ion beside one Cl^- ion and labels the pair “a molecule of sodium chloride”. Explain why this is a misleading representation of solid NaCl.

Answer

Solid NaCl does not contain discrete NaCl molecules. It contains an extended three-dimensional lattice in which each ion is attracted to several oppositely charged neighbours. The formula NaCl gives the simplest whole-number ratio of ions, not the composition of an individual molecule.

4. Delocalisation of Electrons

4A Easy

What is meant by a delocalised electron? Give one metallic and one covalent example.

Answer

A delocalised electron is not confined to a single atom or bond. Metallic example: valence electrons extending throughout a copper lattice. Covalent example: π electrons distributed around the benzene ring.

4B Medium

Explain why the two C–O bonds in the ethanoate ion have the same length even though a single Lewis structure shows one single bond and one double bond.

Answer

The negative charge and π electron density are delocalised over both oxygen atoms. The actual ion is a resonance hybrid, so both C–O bonds have the same average bond order of approximately 1.5 and therefore the same length.

4C Hard

The three N–O bonds in nitrate have equal lengths intermediate between typical N–O and N=O bonds. Use electron delocalisation to predict the average N–O bond order.

Answer

Each resonance contributor contains one N=O bond and two N–O bonds. The total bond order is 2 + 1 + 1 = 4, distributed over three equivalent bonds, so the average N–O bond order is 4 3 .

5. Dot-and-Cross Diagrams

5A Easy

Draw dot-and-cross diagrams for Cl_2 and $MgCl_2$.

Answer

For Cl_2, each chlorine contributes one electron to a shared pair and retains three lone pairs. For $MgCl_2$, magnesium loses two electrons to form Mg^{2+} and each chlorine gains one electron to form two Cl^- ions, each with a complete octet.

5B Medium

Draw a dot-and-cross diagram showing the formation of NH_4^+ from NH_3 and H^+ , clearly indicating the origin of the bonding electrons.

Answer

The diagram should show nitrogen in NH_3 donating its lone pair to H^+ . Both electrons in the newly formed N–H bond should be marked as originating from nitrogen. The entire structure should be enclosed in brackets with an overall + charge.

5C Hard

Gaseous $AlCl_3$ can dimerise to form Al_2Cl_6. Draw a suitable dot-and-cross representation and explain why coordinate bonds form during dimerisation.

Answer

Each aluminium in monomeric $AlCl_3$ has only six electrons around it and is electron-deficient. During dimerisation, a lone pair from a chlorine atom bonded to one aluminium is donated into an empty orbital on the other aluminium. Two bridging chlorine atoms form coordinate bonds, giving Al_2Cl_6.

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6. Formal Charge

6A Easy

Calculate the formal charge on each atom in NH_4^+ .

Answer

For nitrogen, FC = 5 − (0 + 8/2) = +1. Each hydrogen has formal charge zero, giving the overall charge +1.

6B Medium

Draw the major Lewis structures of the cyanate ion, OCN– . Calculate the formal charges and identify the most important contributor.

Answer

Two important contributors are $^-$O–C≡N with formal charges O = −1, C = 0, N = 0, and O– –C– –N– with O = 0, C = 0, N = −1. The first contributor is generally favoured because the negative charge lies on the more electronegative oxygen, although both contribute.

6C Hard

Several Lewis structures can be proposed for N_2O with atom order N–N–O. Assign formal charges and decide which structures contribute most strongly to the resonance hybrid.

Answer

Important contributors include $N{\equiv}N^+{-}O^-$, with formal charges 0, +1, −1, and $^-N{=}N^+{=}O$, with −1, +1, 0. The first is the stronger contributor because the negative charge lies on oxygen. Structures involving larger charge separation contribute less.

7. Resonance

7A Easy

Explain why carbonate is represented using several resonance structures rather than one fixed Lewis structure.

Answer

No single Lewis structure represents the three identical C–O bonds. The π electrons and negative charge are delocalised across all three oxygen atoms, so the actual ion is a resonance hybrid.

7B Medium

Benzene is sometimes drawn with alternating single and double bonds. Explain why this does not mean that benzene rapidly switches between two different structures.

Answer

The resonance structures are alternative representations of the same electron distribution. Benzene has one stable delocalised π system in which all six C–C bonds are equivalent. It does not physically oscillate between two localised structures.

7C Hard

The C–N bond in an amide is shorter than a normal C–N single bond, and the atoms around nitrogen are approximately planar. Explain both observations using resonance.

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Answer

The nitrogen lone pair overlaps with the carbonyl π system, giving a resonance contributor with C=N character. This gives the C–N bond partial double-bond character, shortening it and restricting rotation. Effective p-orbital overlap requires the nitrogen and neighbouring atoms to be approximately planar.

B. Orbital Overlap

8. Formation of σ and π Bonds

8A Easy

State the number of σ and π bonds in ethane, ethene and ethyne.

Answer

Ethane: 7σ, 0π. Ethene: 5σ, 1π. Ethyne: 3σ, 2π.

8B Medium

Describe the orbital overlaps involved in forming the carbon–carbon bonds in ethene and ethyne.

Answer

Ethene: one σ bond forms by end-on overlap of $sp2$ orbitals and one π bond by sideways overlap of unhybridised p orbitals. Ethyne: one σ bond forms by end-on overlap of sp orbitals and two mutually perpendicular π bonds form by sideways p-orbital overlap.

8C Hard

Allene, $H_2C{=}C{=}CH_2$, contains two adjacent double bonds. Predict the relative orientations of its two terminal CH_2 groups using the orientations of the π orbitals.

Answer

The central carbon is sp-hybridised and has two mutually perpendicular unhybridised p orbitals. Each forms a π bond with one terminal carbon. The two π systems are perpendicular, so the planes of the terminal CH_2 groups are also perpendicular.

9. Relative Strength of σ and π Overlap

9A Easy

Explain why a σ bond is normally stronger than a π bond formed from the same atomic orbitals.

Answer

A σ bond results from direct, end-on overlap along the internuclear axis. This gives greater orbital overlap and more electron density between the nuclei than the sideways overlap forming a π bond.

9B Medium

A C=C bond is stronger than a C–C bond, but not twice as strong. Explain this observation.

Answer

A C=C bond contains one σ bond and one π bond. The π bond is weaker than the σ bond because sideways overlap is less effective. Adding a π bond therefore increases bond strength, but does not double it.

9C Hard

Ethene undergoes addition reactions much more readily than ethane even though the total C=C bond is stronger than the C–C bond. Resolve this apparent contradiction.

Answer

Addition to ethene normally breaks only the weaker π component while retaining the C–C σ bond. The exposed π electron density is also readily attacked by electrophiles. Reactivity therefore depends on the accessibility and weakness of the π bond, not simply on total bond energy.

10. Restricted Rotation Around Double Bonds

10A Easy

Explain why free rotation is possible around the C–C bond in ethane but not around the C=C bond in ethene.

Answer

A σ bond is cylindrically symmetrical, so rotation does not significantly change overlap. Rotation around C=C would misalign the parallel p orbitals and destroy the π overlap, requiring considerable energy.

10B Medium

Draw the geometrical isomers of but-2-ene and explain why they can exist as separate compounds.

Answer

But-2-ene forms cis and trans isomers. Each carbon of the C=C bond has two different substituents, and restricted rotation prevents interconversion without breaking the π bond.

10C Hard

Rotation around the C–N bond in an amide is restricted even though it is usually drawn as a single bond. Explain this using orbital overlap and resonance.

Answer

The nitrogen lone pair overlaps with the carbonyl p orbitals. Delocalisation gives the C–N bond partial double-bond character. Rotation would disrupt p-orbital alignment and reduce resonance stabilisation, so it is energetically hindered.

11. Bond Order, Bond Length and Bond Strength

11A Easy

Rank C–C, C=C and C≡C in order of increasing bond length and increasing bond strength.

Answer

Increasing bond length: $\mathrm{C \equiv C < C = C < C - C}$. Increasing bond strength: $\mathrm{C - C < C = C < C \equiv C}$.

11B Medium

The average C–O bond order in carbonate is $\frac{4}{3}$. Explain how this accounts for its equal C–O bond lengths.

Answer

Delocalisation makes all three C–O bonds equivalent. Each has more bonding electron density than a single bond but less than a double bond, so each length lies between normal C–O and C=O bond lengths.

11C Hard

Rank the expected N–O bond lengths in NO_2^+ , NO_2^- and NO_3^- . Justify your answer by determining the average N–O bond order in each species.

Answer

NO_2^+ : bond order 2. NO_2^- : average bond order 1.5. NO_3^- : average bond order 4/3. Since higher bond order usually means a shorter bond, increasing length is $NO_2^+ < NO_2^- < NO_3^-$.

12. Bond Polarity and Reactivity

12A Easy

Explain why HCl contains a polar bond whereas Cl_2 does not.

Answer

Chlorine is more electronegative than hydrogen, so the bonding pair in HCl is drawn towards chlorine, producing $H^{\delta+} - Cl^{\delta-}$. In Cl_2, both atoms have the same electronegativity, so electrons are shared equally.

12B Medium

Use bond polarity to explain why nucleophiles commonly attack the carbon atom of a carbonyl group rather than its oxygen atom.

Answer

Oxygen draws electron density away from carbon, producing a $C^{\delta+} = O^{\delta-}$ bond. Electron-rich nucleophiles are therefore attracted to the electron-deficient carbonyl carbon.

12C Hard

Acyl chlorides are generally more reactive toward nucleophilic attack than amides. Explain this using both inductive effects and electron delocalisation.

Answer

Chlorine withdraws electron density inductively, making the carbonyl carbon strongly electrophilic. In an amide, the nitrogen lone pair is delocalised into the carbonyl group, reducing the positive character of the carbonyl carbon. Chloride is also a much better leaving group than an amide ion.

C. Molecular Geometry

13. Electron-Domain Repulsion

13A Easy

Use electron-domain repulsion to predict the shapes of CH_4, NH_3 and H_2O.

Answer

CH_4: tetrahedral, 109.5°. NH_3: trigonal pyramidal, approximately 107°. H_2O: bent, approximately 104.5°.

13B Medium

Predict the shape of SF_4 and explain the preferred position of its lone pair.

Answer

SF_4 has five electron domains and a trigonal-bipyramidal electron geometry. Its molecular shape is seesaw. The lone pair occupies an equatorial position because this gives only two 90° interactions rather than three for an axial position.

13C Hard

Predict and compare the shapes of ClF_3, XeF_2 and XeF_4, showing how the electron domains arrange themselves before lone pairs are omitted from the molecular shape.

Answer

ClF_3: five domains, trigonal-bipyramidal electron geometry, two equatorial lone pairs, T-shaped. XeF_2: five domains, three equatorial lone pairs, linear. XeF_4: six domains, octahedral electron geometry, two opposite lone pairs, square planar.

14. Bonding Pairs Versus Lone Pairs

14A Easy

State the number of bonding pairs and lone pairs around nitrogen in NH_3.

Answer

Three bonding pairs and one lone pair.

14B Medium

SO_2 has three electron domains around sulfur but is not trigonal planar as a molecular shape. Explain.

Answer

Two domains are S–O bonding domains and one is a lone pair. The electron-domain geometry is trigonal planar, but when the lone pair is omitted, the molecular shape is bent.

14C Hard

Predict the shape of I_3^- . Explain why three lone pairs occupy equatorial positions in the underlying trigonal-bipyramidal electron-domain arrangement.

Answer

The central iodine has five domains: two bonding domains and three lone pairs. The three lone pairs occupy equatorial positions to minimise 90° interactions. The two I–I bonds remain axial and opposite, giving a linear molecular shape.

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15. Relative Strengths of Electron-Pair Repulsions

15A Easy

State the order of repulsion strength among lone pair–lone pair, lone pair–bond pair and bond pair–bond pair interactions.

Answer

lone pair–lone pair > lone pair–bond pair > bond pair–bond pair.

15B Medium

Explain why the H–N–H angle in NH_3 is smaller than the H–C–H angle in CH_4.

Answer

CH_4 contains four bonding pairs, whereas NH_3 contains three bonding pairs and one lone pair. The lone pair occupies more space and repels bonding pairs more strongly, compressing the H–N–H angle below 109.5°.

15C Hard

The H–O–H angle in water is approximately 104.5°, while the F–O–F angle in OF_2 is slightly smaller. Suggest why replacing H with the more electronegative F changes the bond angle.

Answer

Fluorine draws O–F bonding electron density away from oxygen. The bonding domains occupy less space close to the central oxygen and exert weaker bond pair–bond pair repulsion there. The oxygen lone pairs can therefore compress the O–F bonds slightly more strongly.

16. Shapes and Approximate Bond Angles

16A Easy

State the shapes and approximate bond angles of $BeCl_2$, BF_3, CH_4, NH_3 and H_2O.

Answer

$BeCl_2$: linear, 180°. BF_3: trigonal planar, 120°. CH_4: tetrahedral, 109.5°. NH_3: trigonal pyramidal, about 107°. H_2O: bent, about 104.5°.

16B Medium

Predict the shapes and characteristic bond angles in PCl_5 and SF_6.

Answer

PCl_5: trigonal bipyramidal, with 120° equatorial, 90° axial–equatorial and 180° axial–axial angles. SF_6: octahedral, with 90° and 180° angles.

16C Hard

Predict the molecular shape and approximate bond angles of BrF_5. Explain any departures from ideal angles.

Answer

BrF_5 has six domains: five bonding domains and one lone pair. The electron geometry is octahedral and the molecular shape square pyramidal. Ideal angles are approximately 90° and 180°, but lone pair–bond pair repulsion causes slight distortions.

17. Effects of Multiple Bonds on Geometry

17A Easy

Why is each carbon atom in ethene approximately trigonal planar?

Answer

Each carbon has three electron domains and is *sp*2-hybridised. These domains arrange themselves approximately 120° apart in one plane, allowing the unhybridised p orbitals to overlap sideways.

17B Medium

In methanal, the O–C–H angles are slightly greater than the H–C–H angle. Explain why the C=O electron domain causes this distortion.

Answer

A double bond contains greater electron density than a single bond and repels neighbouring bonding domains more strongly. The C=O domain pushes the C–H bonds closer together, decreasing H–C–H and increasing O–C–H.

17C Hard

Compare the geometries of NO_2^+ , NO_2^- and NO_3^- . Account for their different bond angles using electron-domain repulsion and lone pairs.

Answer

NO_2^+ : two bonding domains and no lone pair on N, linear, 180°. NO_2^- : two bonding domains and one lone pair, bent, less than 120°. NO_3^- : three bonding domains and no lone pair, trigonal planar, 120°.

18. Electron-Pair Geometry Versus Molecular Geometry

18A Easy

State the electron-pair geometry and molecular geometry of NH_3.

Answer

Electron-pair geometry: tetrahedral. Molecular geometry: trigonal pyramidal.

18B Medium

State both geometries for SF_4 and explain why the names are different.

Answer

Electron-pair geometry: trigonal bipyramidal. Molecular geometry: seesaw. The first includes all bonding and lone-pair domains; the second describes only atom positions.

18C Hard

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$XeOF_4$ contains four Xe–F bonds, one Xe=O bond and one lone pair around xenon. Determine its electron-pair geometry and molecular geometry.

Answer

The Xe=O double bond counts as one electron domain, giving six domains in total. The electron-pair geometry is octahedral and the molecular geometry square pyramidal. The oxygen is normally opposite the lone pair, with four fluorines forming the square base.

D. Molecular Polarity

19. Electronegativity and Bond Dipoles

19A Easy

Indicate the direction of the bond dipole in H–F, C–Cl and N–H.

Answer

The dipole points towards the more electronegative atom: $H^{\delta+} \rightarrow F^{\delta-}$, $C^{\delta+} \rightarrow Cl^{\delta-}$, and $H^{\delta+} \rightarrow N^{\delta-}$.

19B Medium

Explain why the C–F bond is more polar than the C–Cl bond even though chlorine is also highly electronegative.

Answer

The electronegativity difference between carbon and fluorine is greater than that between carbon and chlorine. The bonding electron density is therefore distributed more unevenly in C–F.

19C Hard

A bond has length 1.30×10^{-10} m and dipole moment 1.50 D. Estimate the magnitude of the separated charge as a fraction of the elementary charge. Use $1\ \text{D} = 3.336 \times 10^{-30}$ C m.

Answer

Using $\mu = qr$, $q = \frac{1.50(3.336\times10^{-30})}{1.30\times10^{-10}} = 3.85 \times 10^{-20}$ C. Thus $q/e \approx \frac{3.85\times10^{-20}}{1.602\times10^{-19}} \approx 0.24$. The charge separation is about $0.24e$.

20. Vector Cancellation of Bond Dipoles

20A Easy

Explain why CO_2 is non-polar whereas H_2O is polar.

Answer

CO_2 is linear, so its two equal C=O bond dipoles point in opposite directions and cancel. H_2O is bent, so its two O–H dipoles do not cancel and produce a net molecular dipole.

20B Medium

BF_3 and NH_3 both contain three polar bonds. Explain why BF_3 is non-polar but NH_3 is polar.

Answer

BF_3 is trigonal planar and symmetrical, so the three B–F dipoles cancel. NH_3 is trigonal pyramidal because of its lone pair, so the N–H dipoles do not cancel.

20C Hard

The two identical bond dipoles in an idealised bent molecule each have magnitude 1.50 D and form an angle of 119°. Calculate the resultant molecular dipole moment.

Answer

For two equal vectors, $\mu_{res} = 2\mu \cos(\theta/2) = 2(1.50)\cos(59.5°) \approx 1.52$ D .

21. Polar Bonds in Non-Polar Molecules

21A Easy

Give one molecule that contains polar bonds but has no permanent molecular dipole.

Answer

CO_2, BF_3, CCl_4 or XeF_4 are acceptable. In each case, symmetry causes bond dipoles to cancel.

21B Medium

Explain why CCl_4 is non-polar but $CHCl_3$ is polar.

Answer

CCl_4 has four identical C–Cl bonds arranged symmetrically in a tetrahedron, so the dipoles cancel. In $CHCl_3$, replacing one C–Cl bond by C–H destroys the symmetry, leaving a net dipole.

21C Hard

Compare the molecular polarities of cis- and trans-1,2-dichloroethene using vector diagrams of their bond dipoles.

Answer

In the cis isomer, the two C–Cl dipoles lie on the same side and reinforce, giving a net molecular dipole. In the trans isomer, the dipoles oppose and approximately cancel, making it much less polar or effectively non-polar.

22. Molecular Symmetry

22A Easy

Explain how symmetry causes the bond dipoles in BF_3 to cancel.

Answer

The three equal B–F dipoles are arranged 120° apart in a trigonal plane. Their vector sum is zero.

22B Medium

Predict whether CF_4, CF_3Cl and CF_2Cl_2 are polar.

Answer

CF_4: non-polar because it is symmetrical. CF_3Cl: polar because one substituent differs. CF_2Cl_2: polar because the C–F and C–Cl dipoles do not cancel completely.

22C Hard

The three isomers of dichlorobenzene have the same bonds but different arrangements. Predict which isomer should have the smallest molecular dipole moment and justify your answer geometrically.

Answer

Para-dichlorobenzene should have the smallest dipole moment because its two C–Cl bond dipoles point in opposite directions across the ring and cancel by symmetry. The ortho and meta isomers have non-zero resultants.

23. Effect of Lone Pairs on Polarity

23A Easy

Explain why NH_3 has a permanent dipole.

Answer

Nitrogen is more electronegative than hydrogen, so each N–H bond is polar. The lone pair gives NH_3 a trigonal-pyramidal shape, preventing cancellation of the bond dipoles.

23B Medium

Compare the polarity of BF_3 and NF_3, including the influence of the lone pair on nitrogen.

Answer

BF_3 is non-polar because it is trigonal planar and symmetrical. NF_3 is trigonal pyramidal and therefore polar. However, its N–F bond dipoles point away from nitrogen and partially oppose the lone-pair-side dipole, so NF_3 has a relatively small net dipole.

23C Hard

XeF_2 is non-polar whereas ClF_3 is polar, even though both have trigonal-bipyramidal electron-domain arrangements. Explain.

Answer

In XeF_2, three lone pairs are equatorial and the two Xe–F bonds are opposite axial bonds, so their dipoles cancel. In ClF_3, two lone pairs are equatorial, leaving a T-shaped arrangement whose three bond dipoles do not cancel.

E. Intermolecular Forces

24. Instantaneous Dipole–Induced Dipole Forces

24A Easy

Explain how London dispersion forces arise in a non-polar atom or molecule.

Answer

Random electron motion can create a temporary uneven charge distribution. This instantaneous dipole induces a dipole in a neighbouring particle, producing an electrostatic attraction.

24B Medium

Explain why xenon has a higher boiling point than argon.

Answer

Xenon has more electrons and a larger, more polarisable electron cloud. It therefore develops stronger instantaneous and induced dipoles, giving stronger London forces.

24C Hard

F_2, Cl_2, Br_2 and I_2 have the same general molecular shape. Explain their progression from gases to liquid to solid at room temperature.

Answer

Down the group, the molecules contain more electrons and their electron clouds become larger and more polarisable. London forces therefore become progressively stronger. Consequently, F_2 and Cl_2 are gases, Br_2 is a liquid and I_2 is a solid at room temperature.

25. Permanent Dipole–Permanent Dipole Forces

25A Easy

What molecular property is required for permanent dipole–dipole attraction?

Answer

The molecules must possess permanent molecular dipoles produced by polar bonds that do not cancel geometrically.

25B Medium

Propanone has a substantially higher boiling point than propane despite having a similar relative molecular mass. Explain.

Answer

Both substances experience London forces, but propanone also has a strongly polar C=O group and therefore permanent dipole–dipole attractions. More energy is needed to separate propanone molecules.

25C Hard

Cis-1,2-dichloroethene has a higher boiling point than its trans isomer, but the trans isomer may pack more efficiently in a crystal. Predict how their boiling points and melting points might differ.

Answer

The cis isomer has a permanent dipole and is expected to have the higher boiling point. The more symmetrical trans isomer may pack more efficiently in the solid lattice and can therefore have a higher melting point despite its lower boiling point.

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26. Hydrogen Bonding

26A Easy

State the structural requirements for conventional hydrogen bonding between molecules.

Answer

A hydrogen atom must be covalently bonded to a highly electronegative atom, normally N, O or F, and a neighbouring molecule must contain an available lone pair on N, O or F.

26B Medium

Ethanol and dimethyl ether have the same molecular formula. Explain why ethanol has the higher boiling point.

Answer

Ethanol contains an O–H group and can form intermolecular hydrogen bonds. Dimethyl ether has an oxygen lone pair but no O–H bond, so pure ether cannot donate conventional hydrogen bonds to itself. It experiences only dipole–dipole and London forces.

26C Hard

Ortho-nitrophenol can form an intramolecular hydrogen bond, whereas para-nitrophenol more readily forms intermolecular hydrogen bonds. Predict their relative boiling points and water solubilities.

Answer

Ortho-nitrophenol is expected to have the lower boiling point because intramolecular hydrogen bonding reduces intermolecular association. It is also generally less water-soluble because its polar groups are partly tied up internally. Para-nitrophenol forms stronger intermolecular hydrogen-bond networks and interacts more readily with water.

27. Polarisability

27A Easy

Define polarisability.

Answer

Polarisability is the ease with which the electron cloud of a particle can be distorted to produce an induced dipole.

27B Medium

Explain why I_2 is more polarisable than Cl_2.

Answer

I_2 has more electrons and a larger electron cloud. Its outer electrons are farther from the nuclei and more weakly held, so the cloud is more easily distorted.

27C Hard

A large molecule may possess no permanent dipole yet have a higher boiling point than a small polar molecule. Explain how this is possible without claiming that London forces are intrinsically a different kind of electrostatic interaction.

Answer

A large molecule can have a highly polarisable electron cloud and many regions of contact with neighbouring molecules. Its numerous instantaneous dipole–induced dipole attractions may collectively exceed the dipole–dipole attractions between small polar molecules. Both are electrostatic attractions arising from unequal charge distributions.

28. Molecular Size, Surface Area and Shape

28A Easy

Explain why hexane has a higher boiling point than butane.

Answer

Hexane has more electrons, a larger and more polarisable electron cloud, and greater surface area. It therefore experiences stronger London forces.

28B Medium

Rank pentane, 2-methylbutane and 2,2-dimethylpropane by boiling point.

Answer

2,2-dimethylpropane < 2-methylbutane < pentane. Greater branching makes molecules more compact and reduces surface contact, weakening London forces.

28C Hard

Branching normally lowers the boiling point of an alkane but can sometimes increase its melting point. Explain why the effects on boiling and melting need not follow the same trend.

Answer

Boiling mainly depends on intermolecular attraction in the liquid, where greater surface contact usually strengthens London forces. Melting also depends strongly on crystal packing. A compact, highly symmetrical branched molecule may pack efficiently into an ordered lattice and therefore have an unexpectedly high melting point.

29. Intermolecular Versus Intramolecular Bonding

29A Easy

When water boils, are O–H covalent bonds broken? Explain.

Answer

No. Boiling separates intact water molecules by overcoming intermolecular attractions, mainly hydrogen bonds. The intramolecular O–H covalent bonds remain intact.

29B Medium

Distinguish the energy changes involved in melting NaCl from those involved in decomposing NaCl into neutral sodium and chlorine atoms.

Answer

Melting partially overcomes ionic lattice attractions so that Na^+ and Cl^- ions become mobile, but the ions remain charged. Producing neutral atoms would require complete lattice separation and electron transfer from Cl^- back to Na^+ , involving fundamentally different and much larger energy changes.

29C Hard

A student says that diamond has a high boiling point because it has "very strong intermolecular forces". Correct this explanation and distinguish clearly between a molecular substance and a network solid.

Answer

Diamond contains no discrete molecules and therefore has no intermolecular forces. Every carbon atom is covalently bonded into an extended three-dimensional network. Vaporisation requires breaking many strong covalent bonds. In a molecular substance, vaporisation normally separates intact molecules by overcoming intermolecular attractions.

30. Solubility and "Like Dissolves Like"

30A Easy

Explain why hexane and water form separate layers.

Answer

Water molecules form strong hydrogen-bonded interactions with one another. Hexane cannot replace these with attractions of comparable strength, so mixing is energetically unfavourable and the substances separate.

30B Medium

Ethanol mixes with both water and hexane to some extent. Explain this using different parts of the ethanol molecule.

Answer

The polar O–H group can form hydrogen bonds with water. The non-polar ethyl group can interact with hexane through London forces. Ethanol therefore has both hydrophilic and hydrophobic character.

30C Hard

Compare the likely water solubilities of benzene, phenol and benzene-1,2-diol. Discuss solute–solvent attractions, disruption of existing interactions and the limitations of simply counting hydrogen bonds.

Answer

Benzene is least soluble because it cannot form strong specific interactions with water. Phenol is more soluble because its O–H group can hydrogen-bond with water. Benzene-1,2-diol is generally still more soluble because it has two O–H groups, although exact solubility also depends on crystal-lattice strength, molecular packing and possible intramolecular hydrogen bonding. Counting hydrogen bonds alone is therefore insufficient.

F. Structures of Solids

31. Ionic Lattices

31A Easy

Explain why solid NaCl does not conduct electricity but molten NaCl does.

Answer

In solid NaCl, ions are fixed in lattice positions and cannot carry charge through the solid. When molten, the ions are mobile and migrate under an electric field.

31B Medium

MgO has a much higher melting point than NaCl. Explain using ionic charge, ionic radius and lattice attraction.

Answer

MgO contains Mg^{2+} and O^{2-} ions, whereas NaCl contains singly charged ions. The larger charge product gives much stronger electrostatic attraction. Mg^{2+} is also relatively small, giving shorter ionic separation and further strengthening the lattice.

31C Hard

A highly charged ionic solid can have a very large lattice enthalpy but still dissolve poorly in water. Explain why lattice enthalpy alone cannot determine solubility.

Answer

Dissolution depends on the total Gibbs energy change. Energy is required to separate ions from the lattice, but energy is released when ions are hydrated, and entropy also contributes. Dissolution is favourable only when the combined hydration and entropy effects compensate sufficiently for lattice separation.

32. Metallic Lattices

32A Easy

Use metallic bonding to explain electrical conductivity and malleability.

Answer

Delocalised electrons can move through the lattice and carry charge. Metallic bonding is non-directional, so layers of metal ions can slide while remaining attracted to the mobile electron cloud, making metals malleable.

32B Medium

Magnesium generally has a higher melting point than sodium. Explain using the charge and size of the metal ions and the number of delocalised electrons.

Answer

Magnesium forms smaller Mg^{2+} ions and contributes two delocalised electrons per atom, whereas sodium forms Na^{+} and contributes one. The attraction between Mg^{2+} ions and the electron sea is therefore stronger.

32C Hard

Pure copper is softer than brass. Explain how differently sized atoms in an alloy interfere with movement of layers while retaining metallic conductivity.

Answer

Zinc atoms have a different size from copper atoms and distort the regular lattice. This makes it harder for layers to slide, increasing hardness. Delocalised electrons remain mobile throughout the alloy, so electrical conductivity is retained, although often reduced.

33. Simple Molecular Solids

33A Easy

Explain why iodine forms a solid that melts at a relatively low temperature and does not conduct electricity.

Answer

Solid iodine consists of discrete I_2 molecules held together by London forces. These are much weaker than covalent or ionic bonds, so relatively little energy is required to melt the solid. It lacks mobile ions or delocalised electrons and therefore does not conduct.

33B Medium

Compare the structures and melting behaviour of solid CO_2 and SiO_2.

Answer

Solid CO_2 is a simple molecular solid containing discrete molecules held by London forces, so it sublimes or melts at relatively low temperature. SiO_2 is a giant covalent network in which strong Si–O bonds extend throughout the solid, giving a very high melting point.

33C Hard

Buckminsterfullerene, C_{60}, is composed only of carbon but has a much lower melting or sublimation temperature than diamond. Explain.

Answer

C_{60} consists of discrete covalent molecules. Strong C–C bonds lie within each molecule, but different molecules are held mainly by London forces. Diamond is a continuous covalent network, so separating it requires breaking strong covalent bonds.

34. Giant Covalent Structures

34A Easy

Explain why diamond is hard and has a very high melting temperature.

Answer

Each carbon atom is covalently bonded to four others in a rigid three-dimensional network. Deformation or melting requires breaking many strong, directional covalent bonds.

34B Medium

SiO_2 and CO_2 contain bonds between elements from the same groups but form very different solids. Explain why.

Answer

Carbon forms strong C=O double bonds through effective p-orbital overlap, allowing stable discrete CO_2 molecules. Silicon has poorer p–p overlap with oxygen and instead forms four strong Si–O single bonds in an extended network.

34C Hard

Silicon and diamond both possess extended covalent structures, but silicon is a semiconductor while diamond is normally an electrical insulator. Suggest a bonding-based explanation.

Answer

Silicon has longer, weaker bonds and a smaller energy gap between its filled valence band and empty conduction band. Thermal energy can promote some electrons into mobile conducting states. Diamond has a much larger band gap, so ordinary thermal energy cannot produce significant charge carriers.

35. Hydrogen-Bonded Solids

35A Easy

Describe the role of hydrogen bonding in holding an ice crystal together.

Answer

Each water molecule can donate two and accept two hydrogen bonds, creating an extended hydrogen-bonded network that maintains the ordered crystal structure.

35B Medium

Explain why ice forms an open lattice and is less dense than liquid water.

Answer

Hydrogen bonding favours an approximately tetrahedral arrangement around each water molecule. This creates an open structure containing empty space. On melting, part of the structure collapses and molecules pack more closely, increasing density.

35C Hard

Urea forms a hydrogen-bonded molecular solid with a relatively high melting point. Explain why it is nevertheless classified as a molecular solid rather than a giant covalent structure.

Answer

Urea contains discrete molecules with strong covalent bonds internally. Separate molecules are connected by hydrogen bonds, not continuous covalent bonds. Its classification depends on the nature of the lattice, not merely on having a relatively high melting point.

36. Structure–Property Relationships

36A Easy

For NaCl, copper, iodine and diamond, state the dominant bonding or attractive force and whether each conducts electricity as a solid.

Answer

NaCl: ionic bonding, does not conduct as a solid. Copper: metallic bonding, conducts. Iodine: London forces between I_2 molecules, does not conduct. Diamond: giant covalent bonding, does not conduct.

36B Medium

An unknown solid has a high melting point, is brittle, dissolves in water and conducts only when molten or dissolved. Deduce its likely structure.

Answer

It is most likely ionic. Strong ionic attractions account for the high melting point, displacement of ion layers accounts for brittleness, and mobile ions conduct in molten or aqueous states.

36C Hard

Three unknown solids have the following properties: P has a low melting point, is volatile and non-conducting; Q has a very high melting point, is hard and conducts as a solid; R has a high melting point, is brittle and conducts only when molten. Propose possible structures for P, Q and R, and identify what additional evidence would remove ambiguity.

Answer

P is probably simple molecular. Q could be a metal or a conducting giant covalent structure. R is probably ionic. Malleability would help distinguish a metal from a brittle network solid; solubility, conductivity of solutions and structural analysis would give further evidence.

37. Graphite, Diamond and Unfamiliar Allotropes

37A Easy

Compare the bonding and electrical conductivity of graphite and diamond.

Answer

Diamond contains four covalent bonds per carbon in a three-dimensional network and has no mobile electrons, so it does not conduct. Graphite contains three covalent bonds per carbon in layers, leaving one electron per carbon delocalised within each layer, so it conducts along the layers.

37B Medium

Explain why graphite is soft even though the covalent bonds within each carbon layer are strong.

Answer

The layers are held together only by relatively weak London forces. They can slide over one another easily even though covalent bonding within each layer is strong.

37C Hard

An unfamiliar carbon allotrope consists of covalently bonded two-dimensional sheets in which every carbon forms three σ bonds and contributes one electron to a delocalised π system. Predict its conductivity, mechanical strength within a sheet and ease of separating the sheets.

Answer

It should conduct within each sheet because of delocalised π electrons. It should be mechanically strong within each sheet because of the extended covalent network. If sheets are held together only by weak attractions, they should be relatively easy to separate or slide.

G. Integrated Problem Types

38. Boiling-Point Ranking

38A Easy

Rank CH_4, NH_3 and H_2O by boiling point and explain the main intermolecular forces present.

Answer

$CH_4 < NH_3 < H_2O$. Methane has only weak London forces. Ammonia forms hydrogen bonds, while water forms a more extensive hydrogen-bonded network because each molecule can donate two and accept two hydrogen bonds.

38B Medium

Rank butane, propanone, propan-1-ol and ethanoic acid by expected boiling point.

Answer

butane < propanone < propan-1-ol < ethanoic acid. Butane has only London forces; propanone also has dipole– dipole attractions; propan-1-ol hydrogen-bonds; ethanoic acid forms particularly stable hydrogen-bonded associations.

38C Hard

Rank the boiling points of pentane, 2-methylbutane, butan-1-ol, ethoxyethane and butanoic acid. Explain every adjacent comparison rather than invoking only relative molecular mass.

Answer

A reasonable ranking is 2-methylbutane < ethoxyethane $\lesssim$ pentane < butan-1-ol < butanoic acid. Branching weakens surface contact; ethoxyethane has a dipole but cannot donate hydrogen bonds; butan-1-ol hydrogen-bonds; butanoic acid forms especially strong hydrogen-bonded associations. The order of ethoxyethane and pentane is close and may require data.

39. Structure from Physical-Property Data

39A Easy

A substance has a low melting point, does not conduct in any state and dissolves in hexane. Suggest its likely structure.

Answer

It is probably a non-polar simple molecular substance. Weak intermolecular forces explain the low melting point, and the absence of mobile charged particles explains the lack of conductivity.

39B Medium

A solid is soft, electrically conducting and composed entirely of carbon. Identify a likely allotrope and explain its properties.

Answer

Graphite is likely. Delocalised electrons within its layers conduct electricity, while weak attractions between layers allow them to slide, making graphite soft.

39C Hard

A colourless solid melts at 780°C, dissolves in water, does not conduct as a solid and produces a conducting solution. Deduce what can and cannot be concluded about its bonding.

Answer

The evidence strongly suggests an ionic solid: high melting point, immobile ions in the solid and mobile ions in solution. However, the precise ions and identity cannot be established from these data alone; some molecular substances ionise in water, so confirmatory evidence may still be needed.

40. Comparing Isomers

40A Easy

Which has the higher boiling point: pentane or 2,2-dimethylpropane?

Answer

Pentane. Its less compact shape gives a greater surface area of contact and stronger London forces.

40B Medium

Explain why straight-chain alkanes generally have higher boiling points than their more highly branched isomers.

Answer

Straight-chain molecules can make more extensive contact with neighbouring molecules, increasing cumulative London attractions. Branching makes molecules more compact and reduces effective surface contact.

40C Hard

One highly symmetrical branched isomer has a lower boiling point but a higher melting point than a less symmetrical straight-chain isomer. Explain both trends.

Answer

Branching reduces surface contact in the liquid, weakening London forces and lowering boiling point. However, high symmetry can allow the branched molecules to pack efficiently into a stable crystal lattice, raising the melting point.

41. Identifying All Intermolecular Forces

41A Easy

Identify all intermolecular forces between HCl molecules.

Answer

HCl molecules experience London forces and permanent dipole–dipole attractions. They do not exhibit conventional hydrogen bonding.

41B Medium

Identify all intermolecular forces present in pure ethanol. Explain why London forces should not be omitted.

Answer

Ethanol exhibits hydrogen bonding, permanent dipole–dipole attractions and London forces. London forces occur in all atoms and molecules because all electron clouds undergo instantaneous fluctuations.

41C Hard

Identify and compare the interactions present in pure ethanoic acid, ethanoic acid dissolved in water and gaseous ethanoic acid dimers.

Answer

Pure ethanoic acid: London forces, dipole–dipole attractions and extensive hydrogen bonding. In water: hydrogen bonding between acid and water, plus London and dipole interactions; some acid ionises. Gas-phase dimers: two acid molecules form a cyclic dimer held by two hydrogen bonds.

42. Predicting Polarity from Unfamiliar Structures

42A Easy

Predict whether SO_3 is polar.

Answer

SO_3 is non-polar. It is trigonal planar and symmetrical, so its three S–O bond dipoles cancel.

42B Medium

Predict whether SF_4 and XeF_4 are polar.

Answer

SF_4 is polar because its seesaw shape is asymmetrical. XeF_4 is non-polar because its square-planar bond dipoles cancel.

42C Hard

A molecule AX_4Y has trigonal-bipyramidal electron geometry, with Y occupying either an axial or equatorial site. Determine which arrangement is likely to possess the larger molecular dipole and what further information is needed.

Answer

Geometry alone is insufficient. In an ideal vector model, replacing one X bond by Y gives a resultant related mainly to the difference between A–X and A–Y bond dipoles. Actual values require bond-dipole magnitudes, bond lengths and information about geometric distortion and any lone-pair effects.

43. Anomalous Properties of Water and Ice

43A Easy

Explain why water has a much higher boiling point than H_2S.

Answer

Water forms strong, extensive hydrogen bonds because oxygen is small and highly electronegative. H_2S does not form comparably strong conventional hydrogen bonds and is held together mainly by weaker dipole and London forces.

43B Medium

Explain why ice floats on liquid water.

Answer

Ice has an open tetrahedral hydrogen-bonded lattice with more empty space than liquid water. Its mass per unit volume is therefore lower, so it floats.

43C Hard

Water reaches maximum density near 4°C, rather than at its freezing point. Propose a molecular explanation involving competition between thermal contraction and hydrogen-bonded structural organisation.

Answer

As water cools from higher temperatures, reduced molecular motion allows molecules to move closer together, increasing density. Below about 4°C, increasingly ordered tetrahedral hydrogen-bonded arrangements create open structures. This structural expansion outweighs ordinary thermal contraction, so density decreases as freezing is approached.

44. Bond-Length and Bond-Energy Comparisons

44A Easy

Compare the bond lengths and strengths of C–C, C=C and C≡C.

Answer

Length: C ≡ C < C = C < C − C. Strength: C − C < C = C < C ≡ C.

44B Medium

Explain why the C–O bonds in carbonate are shorter than a normal C–O single bond but longer than a C=O double bond.

Answer

The π electron density is delocalised over all three C–O bonds. Each has average bond order 4/3, intermediate between single and double, so its length is also intermediate.

44C Hard

Arrange the N–O bonds in NO^+ , NO_2^- and NO_3^- by expected length. State clearly any assumptions used.

Answer

Approximate bond orders are 3 for NO^+ , 1.5 for NO^{2-} , and 4/3 for NO^{3-} . Assuming bond length decreases as bond order increases, increasing length is $NO^+ < NO_2^- < NO_3^-$.

45. Testing Proposed Lewis Structures and Shapes

45A Easy

A student proposes that CO_2 is bent because each oxygen has two lone pairs. Identify the error.

Answer

VSEPR shape is determined by electron domains around the central atom, not lone pairs on terminal atoms. Carbon has two bonding domains and no lone pairs, so CO_2 is linear.

45B Medium

A student proposes a trigonal-planar shape for NH_3. Explain why this is not the molecular geometry.

Answer

Nitrogen has four electron domains: three bonding pairs and one lone pair. The electron-domain geometry is tetrahedral and the molecular geometry is trigonal pyramidal.

45C Hard

A proposed structure for XeF_4 is tetrahedral because xenon forms four bonds. Use electron counting and electron-domain geometry to evaluate the proposal.

Answer

Xenon has eight valence electrons. After forming four Xe–F bonds, two lone pairs remain on xenon, giving six electron domains and an octahedral electron-domain geometry. The two lone pairs occupy opposite positions, leaving a square-planar molecule rather than a tetrahedral one.

Olympiad-Style Extensions

The following problems focus on critical thinking rather than specialised prior knowledge.

Extension 1: Counting Particles in a Unit Cell

A crystalline solid contains two elements, A and B.

- A atoms occupy all eight corners of a cubic unit cell.
- B atoms occupy the centres of all six faces.

1(a) Effective number of A atoms

Calculate the effective number of A atoms in one unit cell.

Answer

Each corner atom is shared by eight neighbouring unit cells, so each contributes $\frac{1}{8}$.

$$N_A = 8\left(\tfrac{1}{8}\right) = 1.$$

Hence,

$$\boxed{N_A = 1.}$$

1(b) Effective number of B atoms

Calculate the effective number of B atoms in one unit cell.

Answer

Each face-centred atom is shared by two adjacent unit cells, so each contributes $\frac{1}{2}$.

$$N_B = 6\left(\tfrac{1}{2}\right) = 3.$$

Hence,

$$\boxed{N_B = 3.}$$

Damien Chemistry Olympiad Notes

1(c) Empirical formula

Determine the empirical formula of the solid.

Answer

The ratio of A to B atoms is

$$A : B = 1 : 3.$$

Therefore, the empirical formula is

$$\boxed{\mathrm{AB_3}.}$$

The formula is not $\mathrm{A_8B_6}$ because atoms on the boundaries of a unit cell are shared with neighbouring cells.

1(d) Recognising a cubic structure

Suppose all A and B atoms were instead atoms of the same element X. How many X atoms would be present effectively in the unit cell, and what type of cubic structure would this represent?

Answer

The total effective number of atoms is

$$1 + 3 = 4.$$

Atoms at the corners and face centres form a face-centred cubic structure. Thus,

$$\boxed{\text{4 atoms per unit cell; face-centred cubic}}$$

1(e) Coordination beyond one unit cell

A student says that the corner A atoms are bonded only to the three B atoms on the three visible faces meeting at that corner. Explain why this conclusion is incorrect.

Answer

A unit cell is only one repeating portion of an extended crystal. The corner atom is also surrounded by atoms in neighbouring unit cells that are not shown inside the selected cube. Coordination must therefore be determined using the full repeating lattice, not one isolated unit cell.

Extension 2: Identifying a Unit Cell Using Density

A metal X forms a cubic crystal. Its unit-cell edge length is

$$a = 400 \ \text{pm}.$$

The metal has molar mass

$$M = 63.5 \ \text{g mol}^{-1}$$

and measured density

$$\rho = 6.59 \ \text{g cm}^{-3}.$$

Possible structures are:

Structure	Effective atoms per unit cell
Simple cubic	1
Body-centred cubic	2
Face-centred cubic	4

Use

$$N_A = 6.022 \times 10^{23} \ \text{mol}^{-1}.$$

2(a) Converting the unit-cell edge length

Convert the unit-cell edge length into centimetres.

Answer

Since 1 pm $= 10^{-10}$ cm,
400 pm $= 400 \times 10^{-10}$ cm $= 4.00 \times 10^{-8}$ cm.
Thus,

$$\boxed{a = 4.00 \times 10^{-8} \text{ cm}.}$$

2(b) Volume of one unit cell

Calculate the volume of one unit cell.

Answer

For a cube,
$V = a^3 = (4.00 \times 10^{-8})^3 = 6.40 \times 10^{-23}$ cm^3.
Hence,

$$\boxed{V = 6.40 \times 10^{-23} \text{ cm}^3.}$$

2(c) Mass of one unit cell

Calculate the mass of one unit cell.

Answer

Using $\rho = \frac{m}{V}$,
$m = \rho V = (6.59)(6.40 \times 10^{-23}) = 4.22 \times 10^{-22}$ g.
Hence,

$$\boxed{m_{\text{cell}} = 4.22 \times 10^{-22} \text{ g}.}$$

2(d) Effective number of atoms

Calculate the effective number of X atoms in each unit cell.

Answer

The mass of one X atom is
$m_X = \frac{63.5}{6.022 \times 10^{23}} = 1.054 \times 10^{-22}$ g.
Therefore,

$$Z = \frac{4.22 \times 10^{-22}}{1.054 \times 10^{-22}} \approx 4.00.$$

Thus,

$$\boxed{Z = 4.}$$

2(e) Identifying the crystal structure

Identify the crystal structure.

Answer

A face-centred cubic unit cell contains four atoms effectively. Therefore, X has a

$$\boxed{\text{face-centred cubic structure.}}$$

2(f) Why a small whole number is expected

Explain why the calculated number of atoms should be close to a small whole number.

Answer

A crystal is formed by repeating identical unit cells. Boundary sites make fixed fractional contributions such as $\frac{1}{8}$, $\frac{1}{2}$, or 1, so common cubic structures give simple whole-number totals.
A value far from a whole number may indicate experimental uncertainty, vacancies, partial occupation, or an incorrect structural model.

Extension 3: Interpreting Bond Potential-Energy Curves

The graph represents the potential energy of two pairs of bonded atoms, P and Q, as a function of internuclear distance. The potential energy is defined to approach zero as the atoms become infinitely separated.

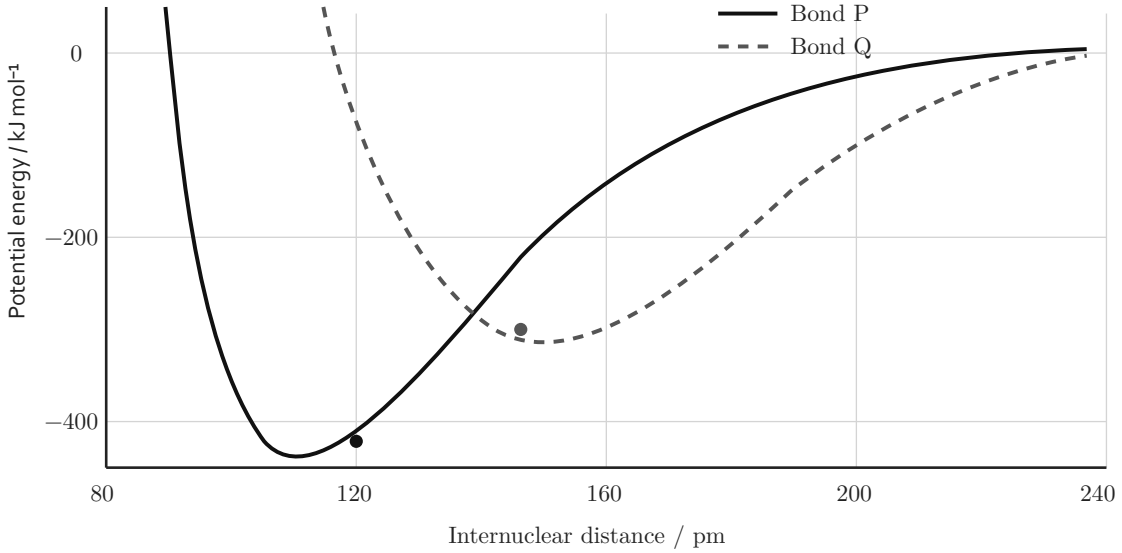


Figure 1: Potential-energy curves for bonds P and Q.

Bond	Distance at energy minimum	Minimum potential energy
P	120 pm	-440 kJ mol^{-1}
Q	155 pm	-250 kJ mol^{-1}

3(a) Meaning of the energy minimum

What does the distance at the minimum of each curve represent?

Answer

The minimum represents the equilibrium bond length: the separation at which the bonded system has minimum potential energy and the net force is zero. Therefore,

$$r_P = 120 \text{ pm}, \quad r_Q = 155 \text{ pm}.$$

3(b) Comparing bond lengths

Which bond is shorter?

Answer

Bond P has the smaller equilibrium internuclear distance, so

P is shorter.

3(c) Comparing bond strengths

Which bond is stronger?

Answer

The depth of the potential-energy well approximates the energy needed to separate the bonded atoms. Bond P requires about 440 kJ mol^{-1}, compared with about 250 kJ mol^{-1} for Q. Thus,

P is stronger.

3(d) Short-range repulsion

Explain why the potential energy rises very steeply when the atoms are brought much closer together than the equilibrium bond length.

Answer

At very short distances, nucleus–nucleus and electron–electron repulsions become very large, and occupied electron clouds are forced to overlap strongly. These repulsive effects dominate and cause the potential energy to rise steeply.

3(e) Large-separation limit

At a distance much greater than the equilibrium bond length, why does the potential energy approach zero?

Answer

At very large separation, the interaction between the atoms becomes negligible and they behave as independent particles. The energy of infinitely separated atoms is chosen as the zero reference.

3(f) Stretching vibrational frequency

Which bond would probably have the higher stretching vibrational frequency, assuming the bonded atoms have similar masses?

Answer

Bond P. Its potential well is steeper near the minimum, corresponding to a larger force constant. A stiffer bond vibrates at a higher frequency when the reduced masses are similar.

3(g) Equilibrium distance and vibration

A student concludes that the atoms in bond P are stationary at exactly 120 pm. Explain why this is not strictly correct.

Answer

The value 120 pm is the equilibrium separation. Bonded atoms vibrate about this distance rather than remaining fixed. Quantum-mechanical zero-point vibration persists even at very low temperature.

Extension 4: Interpreting a Boiling-Point Graph

The approximate boiling points of several hydrides are shown below.

Molecule	Relative molecular mass	Boiling point / K
CH_4	16	112
SiH_4	32	161
GeH_4	77	185
SnH_4	123	221
NH_3	17	240
H_2O	18	373

Molecule	Relative molecular mass	Boiling point / K
HF	20	293

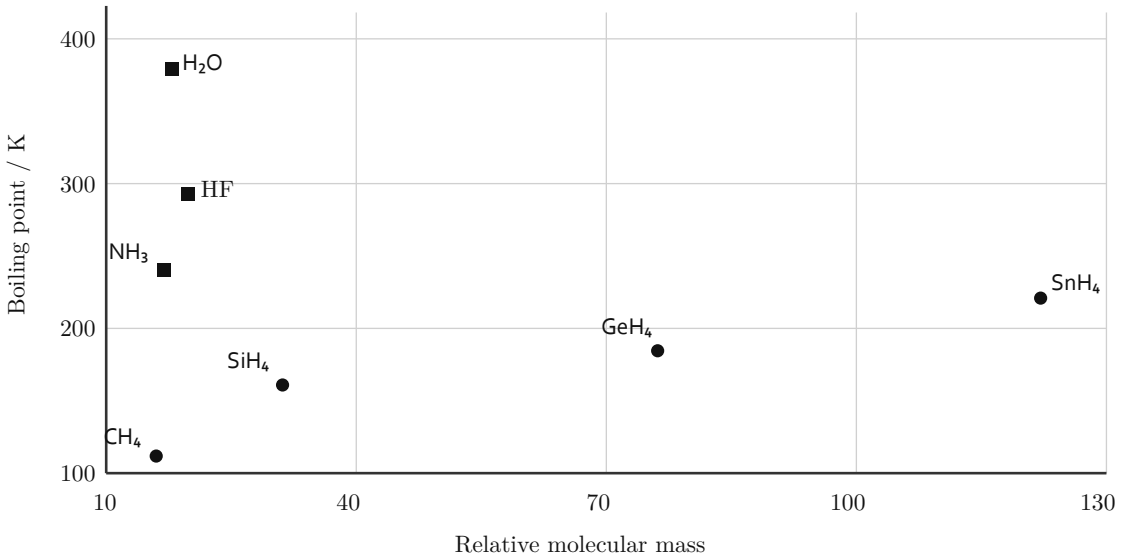


Figure 2: Boiling point against relative molecular mass for selected hydrides.

4(a) Trend among Group 14 hydrides

Describe the trend shown by CH_4, SiH_4, GeH_4, and SnH_4.

Answer

The boiling point increases with relative molecular mass:

$$CH_4 < SiH_4 < GeH_4 < SnH_4.$$

Larger molecules contain more electrons and have more polarisable electron clouds, producing stronger London dispersion forces.

4(b) Why "larger molecules" is incomplete

Why is it incomplete to say only that "larger molecules have stronger intermolecular forces"?

Answer

Size is not the direct microscopic cause. More electrons and a larger, more easily distorted electron cloud increase polarisability, which strengthens instantaneous dipole–induced dipole attractions. The explanation should link size to polarisability and London forces.

4(c) Major outliers

Identify the three major outliers from the trend shown by the Group 14 hydrides.

Answer

The major outliers are

NH_3, H_2O, and HF.

Their boiling points are much higher than expected from molecular mass alone.

4(d) Hydrogen bonding and boiling point

Explain why NH_3, H_2O, and HF lie above the general trend.

Answer

Their molecules form intermolecular hydrogen bonds because hydrogen is directly bonded to a small, highly electronegative atom (N, O, or F). These additional strong intermolecular attractions raise the energy required for vaporisation.

4(e) Why water boils above HF

H_2O has a higher boiling point than HF even though the H–F bond is more polar than the O–H bond. Explain.

Answer

Boiling point depends on the overall intermolecular network, not the polarity of one bond. Each water molecule can donate two and accept two hydrogen bonds, producing an extensive three-dimensional network. HF forms strong hydrogen bonds but a less extensive network. Hence water has the higher boiling point.

4(f) Predicting the boiling point of hydrogen sulfide

Predict whether the boiling point of H_2S would be closer to that of H_2O or to the smooth trend shown by the Group 14 hydrides.

Answer

It would be much closer to the ordinary molecular-mass trend. Sulfur is larger and less electronegative than oxygen, so H_2S does not form strong conventional hydrogen bonds. London forces and weaker permanent dipole interactions dominate.

4(g) Do London forces disappear?

A student states, “NH3 has hydrogen bonding, so London forces are absent.” Evaluate this statement.

Answer

The statement is false. London forces occur between all atoms and molecules. NH_3 experiences London dispersion forces, permanent dipole–dipole attractions, and hydrogen bonding. Hydrogen bonding is additional; it does not replace London forces.

Extension 5: Inferring Structure from Conductivity Data

The electrical conductivities of four substances are measured in different states.

Substance	Solid	Molten	Dissolved in water
W	Very low	High	High
X	High	High	Insoluble
Y	Very low	Very low	Very low
Z	High	—	Insoluble

Further observations are:

- W is brittle and has a high melting point.
- X is malleable.
- Y is volatile and has a low melting point.
- Z is very hard and has an extremely high melting point.

5(a) Structure of W

Deduce the likely structure of W.

Answer

W is most likely ionic. Its ions are fixed in the solid, so the solid does not conduct. The ions become mobile when molten or dissolved, so those states conduct. Strong ionic attraction accounts for the high melting point, while displacement of ion layers can bring like charges together and cause brittleness.

5(b) Structure of X

Deduce the likely structure of X.

Answer

X is most likely metallic. Delocalised electrons conduct in both the solid and molten states, and non-directional metallic bonding allows layers to slide, explaining malleability.

5(c) Structure of Y

Deduce the likely structure of Y.

Answer

Y is most likely a simple molecular substance. Weak intermolecular forces account for volatility and a low melting point, while the absence of mobile ions or delocalised electrons explains the very low conductivity.

5(d) A possible structure for Z

Deduce a possible structure for Z.

Answer

Z could be a conducting giant covalent or extended-network solid containing mobile or delocalised electrons. The very high melting point and hardness indicate strong bonding throughout the structure. Additional evidence is needed because the combination of very high hardness and high conductivity is not characteristic of ordinary graphite.

5(e) Limits of conductivity data

Explain why conductivity data alone may not uniquely identify a structure.

Answer

Different structures can show overlapping electrical behaviour. Metals and graphite both conduct as solids; some molecular substances ionise in water; some giant covalent solids are semiconductors; and an insoluble ionic solid may be impossible to test in aqueous solution.
Conductivity should therefore be combined with melting point, hardness, solubility, malleability, and structural evidence.

End of question set